Decontaminated by Whose Standard? Label-Free Measurement of Decontamination Filter Error

Anonymous ACL submission

Abstract

Open post-training projects routinely state that their training data has been decontaminated against public benchmarks, and readers treat that statement as meaning roughly the same thing across projects. Running five such filters, each from its released implementation, over one corpus against one benchmark suite, we find they flag between 0% and 27.5% of the same items. Disagreement alone, however, invites the reply that different tools simply sit at different thresholds.

Our main contribution settles that. Benchmark items published *after* a corpus was assembled cannot appear in it, so any flag against them is a false positive by construction, identifiable without annotation. Using competition sets held after the corpus cutoff as a negative control, we find the two filters that flag more than a handful carry false-positive rates above a tenth, on each of three independent corpora. That is not a threshold preference but a measured error rate, and its proportion is stable across a 31× range of corpus size. We test the control assumption rather than assuming it, and trace every error to the conventional phrasing of whichever genre the benchmark belongs to.

Because the control is label-free, it also tunes. Applied as a sweep it shows that n-gram size trades sensitivity without buying precision, and that the best-performing method in our comparison ships a default threshold that disables it.

1 Introduction

A benchmark score is evidence of capability only if the benchmark was held out. Open post-training projects generally assert that it was: model cards and papers state that training data was decontaminated against the suites on which results are reported. Readers, and downstream leaderboards, treat those assertions as comparable.

They are comparable only if the underlying procedures agree. This paper asks whether they do, and finds that they do not, by a wide margin.

We take five decontamination filters that ship with real released datasets, reimplement each from its own source, and apply all five to a single corpus against a single benchmark suite. The spread is not subtle. On 100,000 training documents and 502 benchmark items, the filters flag 0, 0, 6, 127 and 138 items respectively. Whether a benchmark problem counts as contaminated is, in practice, a function of which tool was run.

That much is only an observation, and it has an easy dismissal: different tools are tuned differently, and a spread of thresholds is what one would expect. Establishing that filters disagree does not establish that any of them is *wrong*.

Answering that has previously required human adjudication, which is expensive and unreliable at exactly the boundary cases that decide it. We propose a cheaper and more decisive test. Competition mathematics produces new benchmarks continuously and with known dates. An item published after a corpus was assembled cannot be present in it. Any filter that flags such an item has made an error that requires no judgement to identify, and the dismissal does not survive it, because a threshold choice does not explain why half of what a filter reports is wrong.

Using this construction we contribute:

- A label-free method for measuring decontamination false-positive rates, using post-cutoff benchmarks as a negative control (§4), with the control assumption tested

rather than assumed (§4.1).

- Measured false-positive rates of 12.9% and 11.1% for the two filters that flag more than a handful, replicated on two further corpora (§10) and stable in proportion across a 31× range of corpus size (§8).

- A mechanistic account of those errors: all of them arise from a small set of conventional competition-mathematics phrases (§11).

- A direct comparison of five published decontamination filters, each run from its released implementation on identical inputs (§6).

- A tuning procedure built on the control set, showing that n-gram size is a sensitivity parameter rather than a precision parameter, and that fuzzy whole-string matching dominates n-gram matching across the low-error region we measured (§12).

We emphasise at the outset that this is not a finding that any project was careless. Every filter we study does what its authors implemented. The problem is that one word describes procedures measuring different things.

2 Related Work

Work on benchmark contamination divides into detection from the model side and inspection from the data side. Model-side methods ask whether a trained model shows evidence of having seen an evaluation item, through membership inference, loss-based tests, or exchangeability arguments (Oren et al., 2024). A systematic review of 55 such studies finds that no method is reliable across contamination regimes (Nourbakhsh et al., 2026), and Zarzecki et al. (2026) report that across 335 evaluations of three leading paradigms only 60% yield correct outcomes, concluding explicitly that statistical detection cannot yet replace transparent data provenance. Shihab et al. (2026) give an information-theoretic account of why, showing that detectability scales with prevalence and sample size such that audit-scale samples are underpowered by one to two orders of magnitude. Contamination detection in reasoning models is separately fragile (Wang et al., 2026), and detectors built on LLM judges inherit contamination problems of their own (Li et al., 2026a, 2025).

This paper works on the provenance side that such conclusions point toward: the training corpus is public, so overlap can be inspected directly. Prior data-side work has generally audited one corpus with one detector and reported a contamination rate (Sainz et al., 2023), and paraphrase-based evasion is known to defeat surface matching (Yang et al., 2023). Our question is prior to those. Before asking how contaminated a corpus is, we ask whether the quantity is well defined, given that the procedures in use disagree. Recent work pursuing statistical guarantees for detection (Zhang et al., 2026) is complementary: it assumes the target quantity is fixed, which is what we test.

A parallel literature examines the reliability of audit pipelines themselves. Li et al. (2026b) give a five-class taxonomy of pipeline failures in perturbation-based validity audits, and Zarzecki et al. (2026) study distribution shift and scale as limits on model-side detection. Both concern audits of models. To our knowledge no prior work runs multiple published decontamination *implementations* over a common corpus and reports the divergence in what they flag.

The artifacts we study are s1 (Muennighoff et al., 2025), Tülu 3 (Lambert et al., 2024), OpenThoughts and OpenThoughts3 (Guha et al., 2025), and Olmo 3 (Team Olmo, 2025), whose data and decontamination code are all public. Our benchmarks come from OlymMATH (Sun et al., 2025) and MathArena (Dekoninck et al., 2026).

3 The Filters

We reimplemented each filter from its released source, or compiled and ran the released binary where one exists. Table 1 summarises what each does, which in two cases differs from what its paper describes.

Three observations follow from reading the sources.

The filters do not implement the same kind of test. Three are single-stage: a match on any shared n-gram, or a whole-

Filter	Method	Tokenizer	Params from
s1	any shared 8-gram	whitespace, lowercased	code
Tulu 3	coverage fraction over Elasticsearch	spaCy / ES analyzer	paper prose only
OpenThoughts-114K	`fuzz.ratio` ≥ 95	none (raw strings)	code
OpenThoughts3	any shared 13-gram	`Qwen/Qwen2-7B-Instruct`	code
`decon` (Olmo 3)	5-gram detect, then cluster scoring	`cl100k`	code

Table 1: The five filters as implemented rather than as described. Two points deserve emphasis: OpenThoughts-114K performs no n-gram matching at all, and Tulu 3's two governing parameters have no defaults anywhere in its released code.

string similarity above a cutoff. `decon` is two-stage, generating candidates by 5-gram overlap and accepting them only if a weighted score exceeding 0.8 survives cluster expansion. Tulu 3 computes a coverage fraction over documents retrieved by Elasticsearch, so its recall is bounded by retrieval.

The tokenizer is where single-stage filters diverge most. s1 splits on whitespace after lowercasing; OpenThoughts3 hardcodes a model tokenizer. Eight whitespace tokens and eight model tokens describe very different amounts of text, so the same nominal n means different things.

In two cases the released code and the written description do not line up. These are observations about artifacts, not about anyone's conduct, and in both cases there is a benign reading.

The decontamination code released with OpenThoughts-114K contains no n-gram logic at all: it calls `process.extract` with `fuzz.ratio` at a cutoff of 95 on raw strings. A 13-gram path does exist in the same repository, but under OpenThoughts3, a later and separate release. A reader who takes "13-gram" as describing the 114K artifact would be mistaken, though the description is accurate for the artifact it belongs to. Distinguishing the two requires reading the repository.

Tülu 3's `--ngram_size` and `--match_threshold` are declared without defaults and fall through as `None`, so the "8-gram" and "$>$ 50% coverage" that its paper states are not recoverable from the code. The values were presumably supplied at run time by a driver that was not released. Either way a third party cannot reproduce the configuration, so we mark Tülu 3's parameters as prose-derived throughout and treat its numbers accordingly.

Neither observation implies the filters were applied carelessly. Both illustrate the paper's point: what "decontaminated" denotes is fixed by an implementation, and an implementation is only as recoverable as what was released alongside it.

4 Label-Free Evaluation via Temporal Controls

Determining whether a flagged item is genuinely contaminated normally requires human adjudication, which is costly and unreliable at the boundary cases that matter most.

Competition mathematics offers an alternative. New problem sets are released continuously, at publicly known dates. For a corpus assembled at time t, any benchmark published after t cannot be contained in it, whatever a filter reports. Flags against such items are false positives by construction: not estimates, not adjudications, but logical impossibilities.

This yields a two-group design. A **measurement set** of benchmarks published before the corpus cutoff, where flags may be correct or incorrect, and a **control set** published after it, where every flag is wrong. The control set gives a false-positive rate directly, with no annotation, and can be recomputed for any corpus whose assembly date is known.

The design has one requirement: the cutoff must be established independently of the filters being evaluated. We use the corpus's public release date as an upper bound, and exclude benchmarks falling near it.

4.1 Testing the assumption

That upper bound is load-bearing, and Dolci is a remix of separately curated corpora, so in principle a component could postdate the release of the whole. We therefore test it rather

than assert it. A benchmark item genuinely present in a corpus shares long spans with some training document; a coincidental match shares only short ones. For every benchmark item we computed the longest n for which it shares an n-gram with any training document.

Longest match	Measure (502)	Control (280)
no match	375	244
≥ 8	91	25
≥ 10	29	9
≥ 13	7	1
≥ 15	0	1

Table 2: Longest word n-gram each benchmark item shares with the corpus. Rows are *disjoint buckets*, keyed on an item's longest match, not cumulative counts. Summing them recovers Table 3: 91+29+7 = 127 measurement and 25+9+1+1 = 36 control items share some 8-gram, which is what the s1 filter reports.

No control item matches at a length that would indicate presence, and inspection of the five longest control spans finds no problem content at all. The longest, at n=15, is *"where m and n are relatively prime positive integers, find $m + n$"*, the conventional closing of an AIME problem. Three of the remaining four are shorter variants of that same closing, and the fifth is *"reads the same from left to right as from right"*, the standard way of stating a palindrome condition. These are genre conventions, not leaked text.

The longest single shared span anywhere in this study therefore belongs to an item that provably postdates the corpus, and consists entirely of boilerplate. That is both the assumption discharged and the clearest single illustration of the mechanism analysed in §11.

5 Experimental Setup

Corpus. `allenai/Dolci-Think-SFT-32B`, the post-training data for Olmo 3 (2,253,684 examples). Olmo 3 was announced 20 November 2025, so we treat the corpus as closed by roughly October 2025. We use a 100,000-document sample for the main results and up to 390,000 for the scaling analysis. Filters read the first user turn of each example, the field their own implementations target.

Dolci is a remix of separately curated corpora. In our sample the largest slices are OpenThoughts3 Math (34,835), Python Algorithms (21,090), Persona Precise IF (10,089), Nemotron Code (5,232) and SYNTHETIC-2 (4,914). Each was decontaminated, if at all, by whoever assembled it.

Filter	Measure (502)	Control (280)
`decon`	0 (0.0%)	0 (0.0%)
fuzzy, cutoff 95	0 (0.0%)	0 (0.0%)
Tülu 3 coverage	6 (1.2%)	0 (0.0%)
s1 8-gram	127 (25.3%)	36 (12.9%)
OpenThoughts3 13-gram	138 (27.5%)	31 (11.1%)

Table 3: Benchmark items flagged, 100,000 Dolci documents. Every control-set flag is a false positive by construction.

Measurement set (502 items). OlymMATH (200) plus ten MathArena competitions held on or before September 2025: Apex 2025, the Apex shortlist, SMT 2025, CMIMC 2025, BRUMO 2025, AIME 2025, HMMT February 2025, both AIME 2024 papers, and HMMT February 2024.

Control set (280 items). Eight MathArena sets from January to August 2026: AIME 2026, HMMT February 2026, USAMO 2026, three ArXivMath sets and two BrokenArXiv sets. All postdate the corpus by at least three months.

6 Filters Disagree

Table 3 gives the central result. The filters separate into two groups with nothing between them. Three flag almost nothing and are never wrong. Two flag a quarter of the benchmark and are wrong about more than a tenth of it. There is no shared standard for where on this range the word "decontaminated" should sit.

Agreement between them is close to absent. Of the 201 items (40.0% of the benchmark) flagged by at least one filter, 133 (66.2%; 95% CI [59.6, 72.6] by bootstrap over benchmark items, B=10,000) are flagged by *exactly one*. Two items reach three of five, and none reaches four. The highest pairwise Jaccard overlap is 0.318, between the two n-gram filters; every other pair falls below 0.05.

`decon`'s zero is robust rather than marginal: it returns zero at its default score threshold of 0.8, zero at 0.6, a single contaminated docu-

ment at 0.4, and 646 only at 0.2, a quarter of the default.

What `decon` was applied to. We run `decon` as a published filter, not as a claim about which data Ai2 filtered with it. The Olmo 3 paper documents the tool under mid-training; Dolci Think SFT lists decontamination as a curation step without naming a tool, and the dataset card describes no procedure. `decon`'s zero on this corpus is therefore what that tool reports, not confirmation that this split was filtered with it. The point generalises: even within a single project the documented filter and the filter applied to a given split need not coincide, and here the latter is not recoverable from what was released.

A caution about partial reimplementation. Our first attempt at `decon` implemented only its candidate-generation stage, a 5-gram match over `cl100k` tokens. That version flagged 199 of 200 items. The real binary flags none. Reporting the approximation would have inverted the result and misrepresented the tool. A filter is its whole pipeline, and two-stage filters cannot be approximated by their first stage.

7 Measured False-Positive Rates

We report two quantities and keep them distinct throughout. The **false-positive rate** is control flags divided by control items: the share of provably-absent items a filter nonetheless flags. The **estimated false-discovery share** is the control rate divided by the measurement rate: an estimate of how much of a filter's output on real data is noise. The first is measured; the second is inferred under the assumption that the two benchmark groups attract spurious matches at similar rates (§9).

Both n-gram filters flag items that cannot be in the corpus. The false-positive rate is 12.9% for s1 (95% CI $[9.3, 16.8]$) and 11.1% for OpenThoughts3 ($[7.5, 15.0]$), against measurement rates of 25.3% ($[21.5, 29.1]$) and 27.5% ($[23.5, 31.7]$). Intervals are bootstrap over benchmark items, B=10,000; the control set has 280 items, so these carry roughly four points of uncertainty and both nonetheless exclude zero by a wide margin.

The estimated false-discovery share follows: $12.9/25.3 = 0.51$ for s1 and $11.1/27.5 = 0.40$ for OpenThoughts3. Roughly half of what s1 reports, and two fifths of what OpenThoughts3 reports, is estimated to be noise. Subtracting leaves 12.4% and 16.4% as estimates of genuine detection. The three conservative filters have no false positives and almost no true positives.

8 The Error Rate Is Intrinsic

A single-corpus measurement invites the objection that it may not survive scaling. It does not need to: Figure 1 shows the error fraction is invariant. Across seven corpus sizes from 12,500 to 390,000 documents, absolute rates rise by factors of 2.4 to 3.2, for false positives as much as for detections, while the *proportion* of error stays flat. s1's false-positive fraction runs 0.49, 0.52, 0.61, 0.51, 0.53, 0.53, 0.54; OpenThoughts3's runs 0.41, 0.36, 0.37, 0.40, 0.40, 0.43, 0.46, drifting upward, so that filter becomes relatively worse as the corpus grows.

The corrected signal does grow, from 7.0% to 12.8% for s1 and 7.6% to 19.7% for OpenThoughts3, so real overlap is found beneath the noise. It is outnumbered at close to a fixed ratio, and no choice of corpus size changes that.

`fuzz.ratio` returns zero at every scale up to 250,000 and one item at 390,000; `decon` returns zero throughout. The architectural disagreement between whole-string and n-gram matching is therefore scale-invariant, and is the more portable of our two findings.

9 A Matched Same-Competition Check

The estimated false-discovery share assumes the two benchmark groups attract spurious matches at comparable rates. That is testable without any new machinery. AIME 2025 and AIME 2026 are the same competition one year apart, written to the same conventions by the same organisers; only the 2025 edition can be inside the corpus. If both flag at the same rate, the measurement-side flags are mostly boilerplate too.

For s1 the paired rates are indistinguishable: 60.0 against 56.7 on AIME, and 20.0 against 21.2 on HMMT, where the control side is in fact the higher of the two. On these two competitions essentially all of s1's flagging is

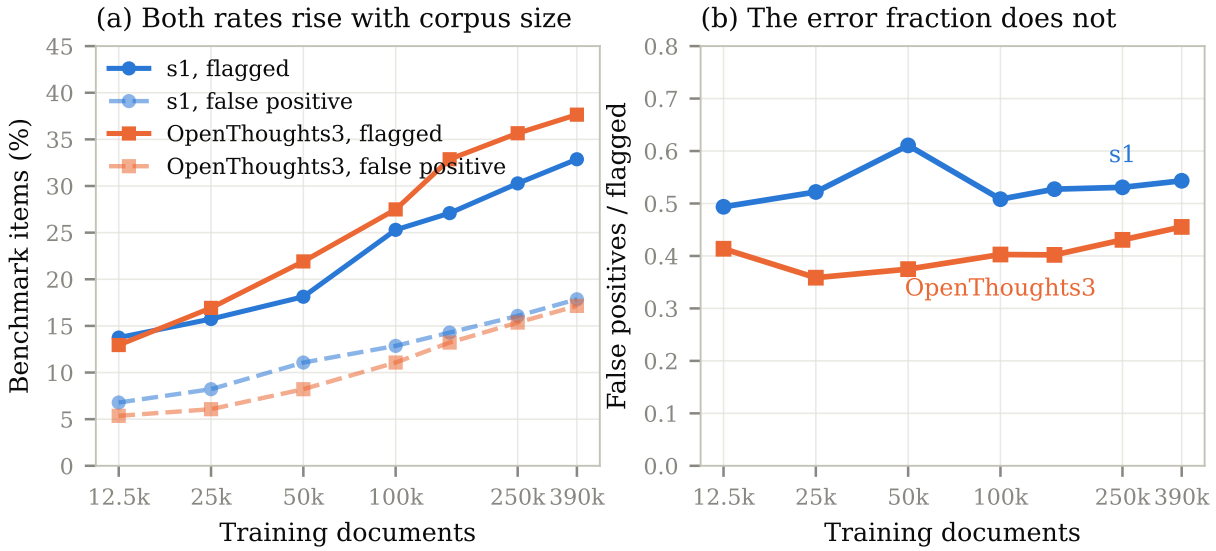


Figure 1: Across a 31× range of corpus size, both the flag rate and the false-positive rate rise by factors of 2.4 to 3.2 (a), while the fraction of each filter's output that is a false positive stays flat (b). The error is a property of the method, not of how much data it is given.

Competition	Group	s1	OT3
AIME 2025	measure	60.0%	53.3%
AIME 2026	control	56.7%	36.7%
HMMT Feb 2025	measure	20.0%	26.7%
HMMT Feb 2026	control	21.2%	12.1%

Table 4: Matched competition pairs one year apart, 100,000 Dolci documents. Only the 2025 editions can be present in the corpus.

explained by shared phrasing rather than by contamination. OpenThoughts3 behaves differently, showing a gap on both pairs (53.3 against 36.7; 26.7 against 12.1), which is what genuine detection above the boilerplate floor looks like.

Two cautions. These are two of the ten measurement competitions, and AIME is the most formulaic of them, so the near-equality for s1 should not be read as holding across the whole measurement set. And the comparison supports the comparable-rate assumption rather than proving it; it is evidence for the inference in §7, not a substitute for it.

10 Two Further Corpora, and a Moving Control

To separate a property of these filters from a property of Dolci, we repeat the measurement on two more corpora. Each has its own release date, so each gets its own split: the control set is a function of the corpus, not a fixed list.

The Tülu 3 SFT mathematics subset shipped in November 2024, so every 2025 competition postdates it and joins the control side, leaving 722 control items and only the three 2024 competitions as measurement. Nemotron-SFT-Math-v4 shipped on 1 May 2026, which falls *between* the two groups used for Dolci: benchmarks published before that date become measurement, and only June 2026 onward can serve as control. Nemotron's later date also forces an arXiv-derived control set, since no competition set after May 2026 was available.

Corpus	Filter	Detect	FP
Tülu 3 (Nov 2024)	s1 8-gram	3.3%	6.2%
	OT3 13-gram	1.7%	6.9%
	fuzzy / cov	0.0%	0.0%
Nemotron (May 2026)	s1 8-gram	15.4%	5.4%
	OT3 13-gram	23.8%	16.1%
	fuzzy / cov	0.1%	0.0%

Table 5: Two further corpora, 100,000 documents each. Tülu 3 has 60 measurement and 722 control items; Nemotron 680 and 261. Each split is derived from that corpus's own release date.

Both n-gram filters again flag items that cannot be present, so the phenomenon holds across all three corpora. We draw no conclusion about detection on Tülu 3, whose 60 measurement items give an estimate with a 95% interval of roughly $[0.4, 11.5]$ percent; its

value here is the large control side. The magnitudes do not transfer, and neither does the ordering. On Dolci s1 has the larger estimated false-discovery share (0.51 against 0.40); on Nemotron the ranking reverses (0.35 against 0.68), and OpenThoughts3's 16.1% is the highest false-positive rate we measure anywhere. No filter is reliably the more precise one, which is the practical form of the paper's claim: an error profile is a property of a filter *and* a corpus together, and cannot be carried over.

A confound we cannot remove. Nemotron's later release date forces a later control set, and the only benchmarks available after May 2026 are arXiv-derived rather than competition mathematics. The cross-corpus comparison of false-positive rates therefore mixes a corpus effect with a benchmark-genre effect, and we cannot separate them with the data available. The within-corpus comparisons between filters are unaffected, since each uses a single control set. The mechanism, however, does carry over. Tracing Nemotron's false positives the same way finds all 14 of them caused by just 22 distinct 8-grams, the same concentration seen on Dolci, and none of them problem content. What differs is the vocabulary: instead of AIME answer formats the recurring spans are research-paper conventions such as *"over an algebraically closed field k of characteristic"*, *"if one can be obtained from the other"* and *"a set of n points in the plane"*. One of the most frequent is not mathematics at all but LaTeX markup, `\begin{pmatrix} a & b \\ c & d`, matched across 20 training documents. The mechanism is therefore general to a genre having conventions; the particular phrases are specific to which genre.

11 Why the Errors Happen

All 36 of s1's false positives trace to 73 distinct 8-grams, and none is a fragment of a leaked problem. They are the conventional phrasing of competition mathematics: *find the sum of all possible values of* (166 documents), *let a and b be positive integers such* (67), *is not divisible by the square of any* (67), *where m and n are relatively prime positive* (8 documents, matching 6 distinct control items).

AIME problems conventionally close by asking for $m + n$ where m and n are relatively prime; "not divisible by the square of any prime" is standard framing for a radical-form answer. These conventions are shared by every problem of the type, including problems written months after the corpus was assembled.

This explains the precision ceiling of §12. Competition mathematics has a fixed inventory of set phrases eight or more words long, so an eight-word window can be filled entirely by convention, carrying no problem-specific content. Increasing n does not escape the problem, because the phrases are themselves long.

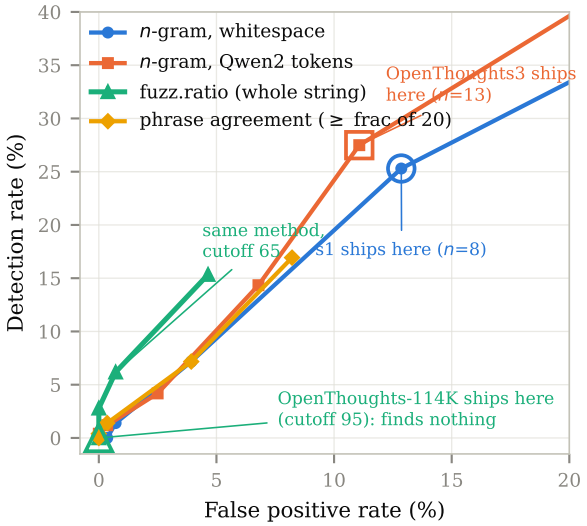


Figure 2: Precision/recall frontier, derived without labels. Rings mark each project's shipped default. Whole-string fuzzy matching dominates both n-gram families in the low-error region, but the project using it ships a cutoff of 95, at which it flags nothing. Points beyond 20% false positives are omitted; at n=5 both n-gram families flag essentially the whole benchmark.

12 Tuning Without Labels

Because control-set flags are provably wrong, sweeping a filter's parameters over both groups yields a precision/recall curve with no annotation anywhere (Figure 2).

***n*-gram size is a sensitivity knob, not a precision knob.** For whitespace tokens the error fraction is pinned near 0.51 from n=8 to n=13 while detection falls from 25.3% to 1.4%: raising n buys no precision, only insensitivity. At n=15 detection reaches zero while false positives persist, so everything found is wrong. Model-token n-grams do improve with

n, from an error fraction of 1.00 at n=5 to 0.40 at n=13 and 0.30 at n=25, but reaching a fraction below 0.1 costs all but 0.4% of detection.

Notably, s1 (whitespace, n=8: 25.3% detection, 12.9% error) and OpenThoughts3 (Qwen, n=13: 27.5%, 11.1%) sit at nearly the same operating point despite sharing no parameters.

The best method ships the worst default. Sweeping `fuzz.ratio` gives 0% detection at its shipped cutoff of 95, 2.8% at 70 with no false positives at all, 6.2% at 65 with 0.7%, and 15.3% at 60 with 4.6%. At comparable recall this dominates n-gram matching: 6.2% detection at 0.7% error against a whitespace 10-gram's 7.2% at 3.9%, a fivefold difference. The method that performs best in our comparison is the one whose default disables it, and the remedy is a threshold roughly thirty points lower.

Where the flagged items come from. Mapping flagged items back to the Dolci slice that triggered them shows the overlap is concentrated rather than spread: SYNTHETIC-2 supplies 4.9% of the corpus sample and accounts for 72 of 138 flagged items, a tenfold enrichment, while Python Algorithms supplies 21.1% and accounts for 3. Appendix A gives the full breakdown and the document-to-item ratio that separates boilerplate from item-level overlap.

13 Testing the Suggested Fix

§11 predicts that requiring agreement across spans spread through an item should suppress single-stock-phrase matches. A filter of that shape is already in use: rather than firing on any shared n-gram, it samples roughly twenty evenly spaced 8-gram phrases per benchmark item and flags only when a fraction of them are present. We implemented it and swept that fraction.

Required	Detect	FP	FD share
≥ 0.50	0.0%	0.0%	–
≥ 0.30	0.4%	0.0%	0.00
≥ 0.20	1.4%	0.4%	0.26
≥ 0.10	7.2%	3.9%	0.55

Table 6: Phrase-agreement filtering on Dolci, 100,000 documents.

The prediction holds at the precise end. At 1.4% detection the phrase-agreement filter makes 0.4% false positives, an estimated false-discovery share of 0.26, against 0.51 for a whitespace 13-gram at the same detection rate: half the error for the same recall. At its shipped setting of one half it flags nothing at all on this corpus, joining the conservative group.

The advantage disappears as the requirement loosens: at ≥ 0.10 the share is 0.55, indistinguishable from a plain n-gram, since two of twenty phrases is barely a constraint. Whole-string matching also still dominates throughout. Requiring phrase agreement is a real improvement over single n-gram matching, but not a substitute for a better-shaped metric.

14 Recommendations

1. **Report the procedure, not the word.** "Decontaminated" is uninformative without the tokenizer, window size, threshold, and benchmark list. Two of our five filters cannot be reproduced from their papers alone.

2. **Publish a temporal control.** Any project can report its false-positive rate against benchmarks postdating its corpus, at no annotation cost.

3. **Prefer whole-string similarity, tuned.** In our comparison fuzzy matching dominated n-gram matching across the low-error region we measured, provided its threshold is calibrated rather than left at 95.

4. **Do not compare contamination rates across projects.** They are not measurements of the same quantity.

15 Conclusion

Five published decontamination filters, run over one corpus against one benchmark suite, flag between 0% and 27.5% of the same items. Measured against benchmarks that postdate the corpus and so cannot be in it, the two that flag more than a handful carry false-positive rates above a tenth, on each of three corpora, at a proportion stable across a 31× range of corpus size. Those errors trace to the conventional phrasing of whichever genre the bench-

mark belongs to, down to LaTeX markup. Neither filter is reliably the more precise one: the ordering reverses between corpora. A decontamination claim is therefore a property of a procedure applied to a corpus, not of the dataset, and the procedures in current use do not agree.

The harness, the filter implementations and the temporal control sets accompany this submission as anonymised supplementary material, so that any project can measure its own false-positive rate without annotating anything.

Limitations

Our own provenance. Every number above was regenerated from committed code before submission rather than carried over from the exploratory scripts that produced it. Three did not survive: a threshold sweep and the slice analysis had been computed against an earlier 200-item benchmark set and then reported alongside the final one, and two error fractions had been formed by dividing rounded percentages rather than counts. All three are corrected here. This is the same drift the paper documents elsewhere, and it was caught only by regenerating from a clean checkout, which is the practice we recommend rather than one we can claim was unnecessary.

We do not measure score inflation. This work measures which items filters flag, not what removing them does to a reported benchmark number. Establishing that would require retraining under each filter, which we did not do.

One domain. All results are on mathematics. The false-positive measurement replicates on a second corpus (§10), but whether the stock-phrase mechanism of §11 generalises to code or natural-language benchmarks is untested, though we would expect analogous conventions. The detection-side results rest on Dolci alone.

The control assumes a known cutoff. We use a public release date as an upper bound on corpus assembly. A corpus updated after release, or one whose components postdate the release of the whole, would violate this. We test the assumption directly in §efsec:validate and find no control item present, but the test is specific to this corpus and must be repeated for any other.

Tulu 3 is evaluated on prose-derived parameters. Its code declares no defaults, so we used the values its paper states. A different reading would give different results, which is itself part of our argument.

`decon` is evaluated on its defaults only. We swept its score threshold but did not explore its other parameters, of which there are many.

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A Slice-Level Breakdown

Mapping flagged items back to the Dolci source that triggered them shows overlap is not spread evenly across the remix. Under the OpenThoughts3 filter, which flags 138 measurement items, SYNTHETIC-2 supplies 4.9% of our corpus sample and accounts for 72 of them, a tenfold enrichment. Nemotron Code supplies 5.2% and accounts for 26 (3.6×). By contrast OpenThoughts3 Math supplies 34.8% of the corpus and accounts for 70 (1.5×), and Python Algorithms supplies 21.1% and accounts for 3 (0.1×).

The ratio of triggering documents to distinct flagged items is itself diagnostic. Python Algorithms triggers 278 documents against only 3 items, the signature of shared boilerplate; SYNTHETIC-2 triggers 109 documents against 72 items, the signature of item-level overlap. Filters reporting only a document count cannot distinguish these, and all five of ours do exactly that.